

Module 4 Portfolio Milestone – Bank Marketing

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Course: MIS450 Data Mining

Introduction

The purpose of this milestone is to explore the Bank Marketing dataset using SAS Studio. The dataset contains 4,119 observations and 21 variables. The overall goal of the project is to predict whether a client will subscribe to a term deposit based on demographic, financial, and marketing campaign information.

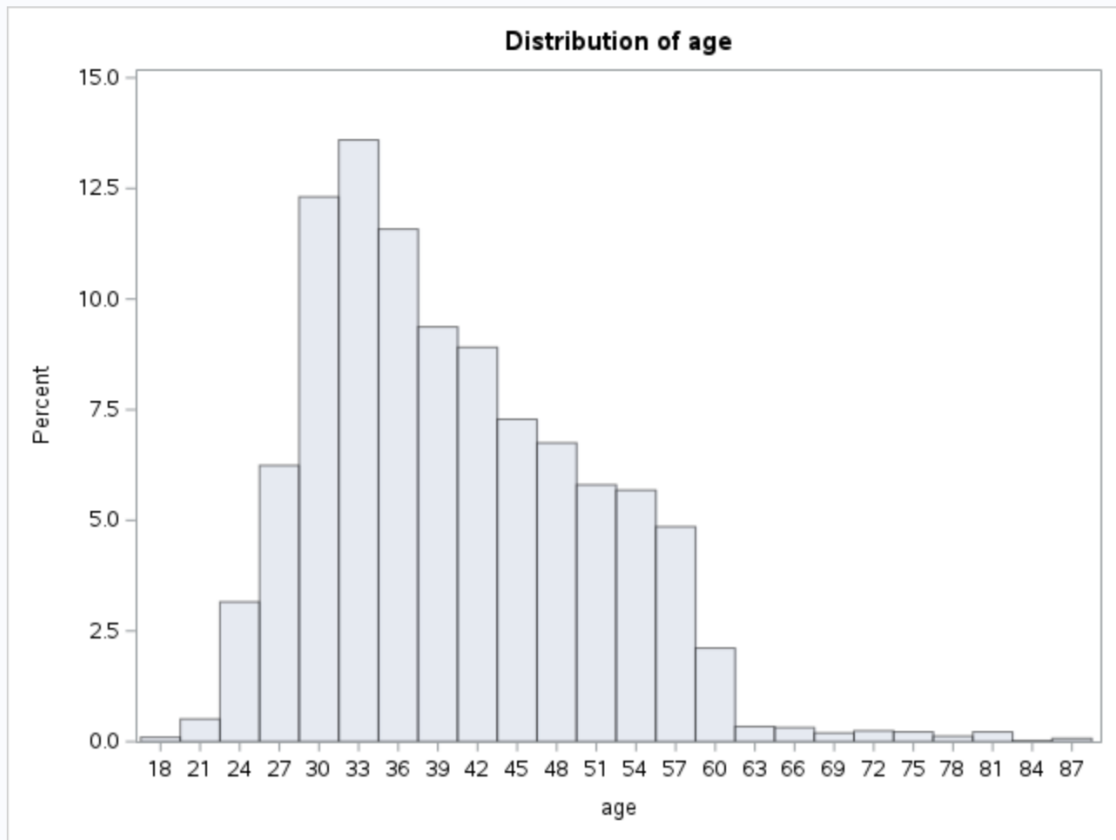
Summary Statistics and Graphical Summaries

The summary statistics show that the average customer is approximately 40 years old, and the average call duration is about 257 seconds. Most customers were contacted only a few times during the campaign, while a small number received many contacts. These statistics provide an overall understanding of the numerical variables before building a predictive model.

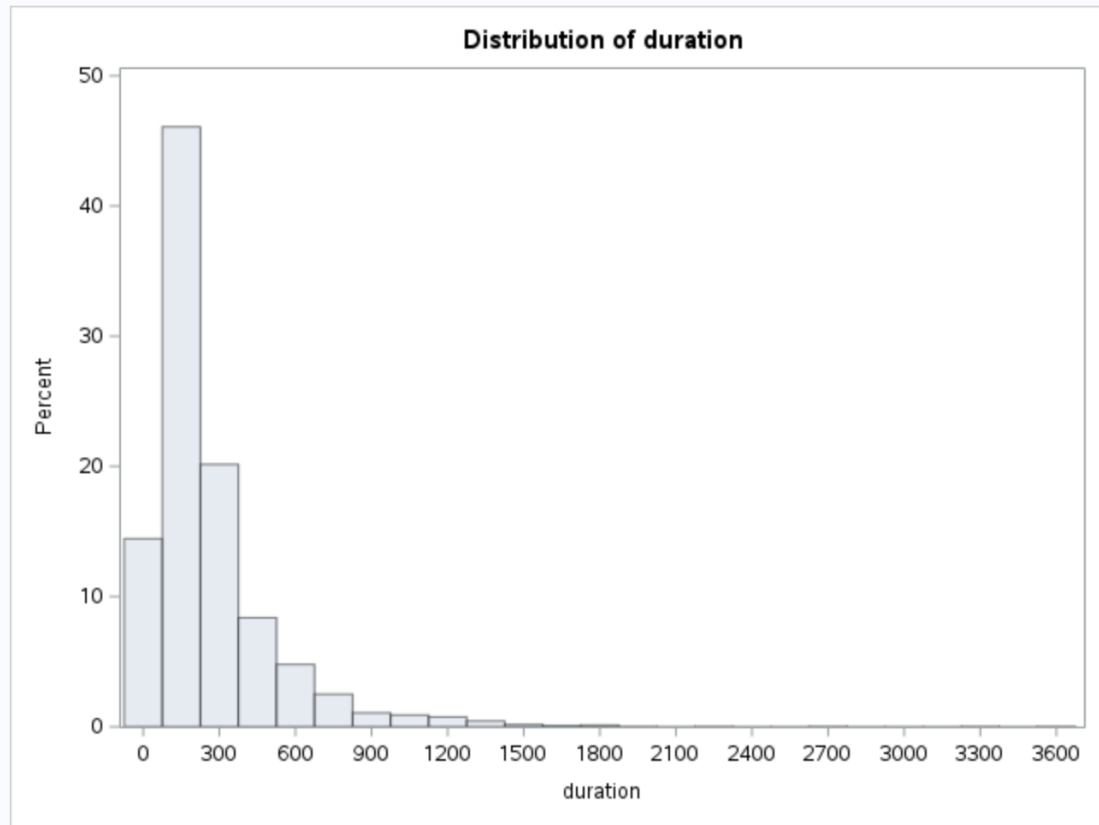
- Summary Statistics

Variable	Mean	Std Dev	Minimum	Maximum	N
age	40.1136198	10.3133615	18.0000000	88.0000000	4119
duration	256.7880554	254.7037361	0	3643.00	4119
campaign	2.5372663	2.5681592	1.0000000	35.0000000	4119
pdays	960.4221899	191.9227858	0	999.0000000	4119
previous	0.1903375	0.5417883	0	6.0000000	4119
emp.var.rate	0.0849721	1.5631145	-3.4000000	1.4000000	4119
cons.price.idx	93.5797043	0.5793488	92.2010000	94.7670000	4119
cons.conf.idx	-40.4991017	4.5945775	-50.8000000	-26.9000000	4119
euribor3m	3.6213557	1.7335912	0.6350000	5.0450000	4119
nr.employed	5166.48	73.6679036	4963.60	5228.10	4119

- Histogram of Age

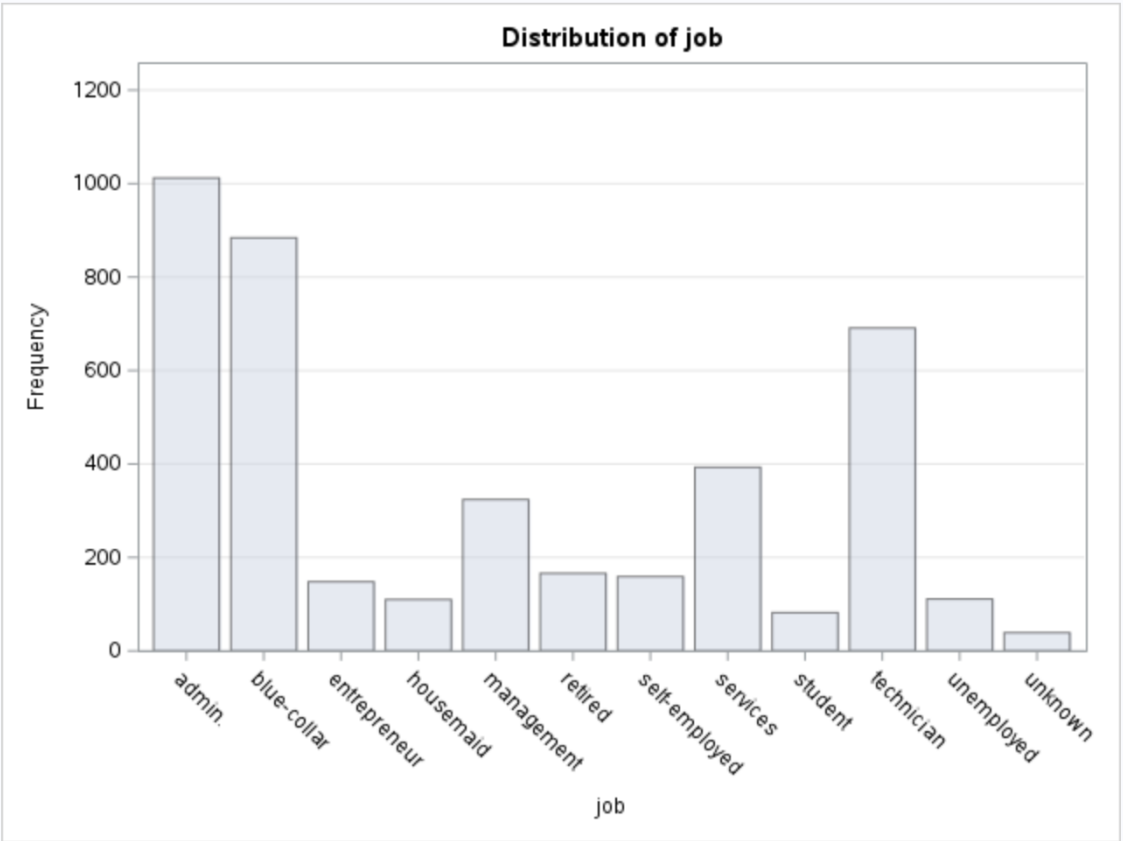


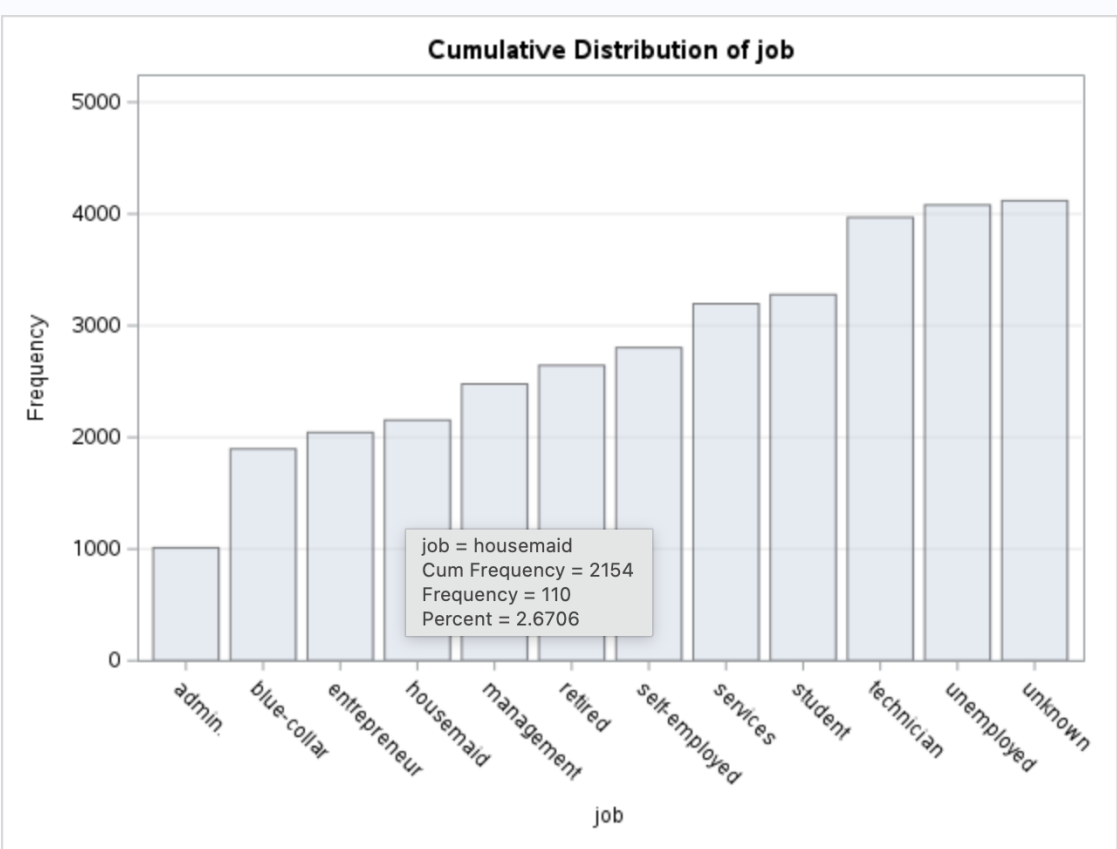
- Histogram of Duration



- Bar Chart of Job

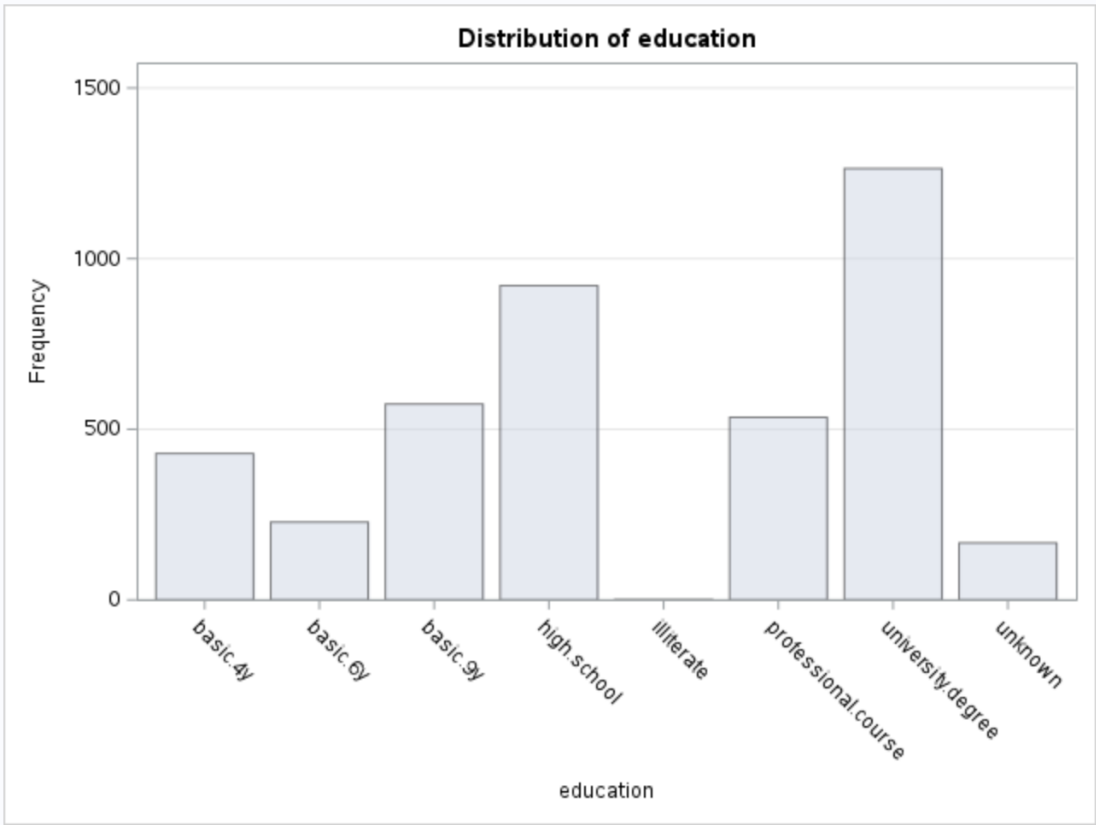
job	Frequency	Percent	Cumulative Frequency	Cumulative Percent
admin.	1012	24.57	1012	24.57
blue-collar	884	21.46	1896	46.03
entrepreneur	148	3.59	2044	49.62
housemaid	110	2.67	2154	52.29
management	324	7.87	2478	60.16
retired	166	4.03	2644	64.19
self-employed	159	3.86	2803	68.05
services	393	9.54	3196	77.59
student	82	1.99	3278	79.58
technician	691	16.78	3969	96.36
unemployed	111	2.69	4080	99.05
unknown	39	0.95	4119	100.00

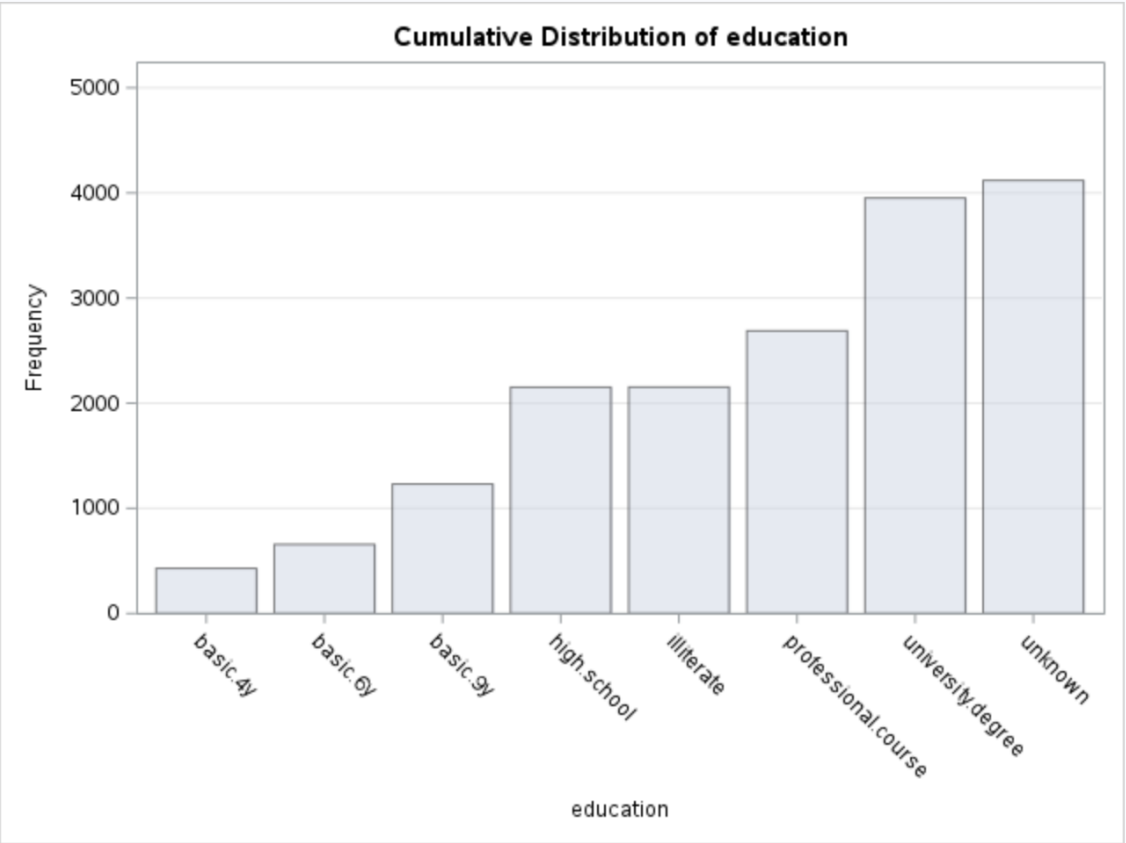




- Bar Chart of Education

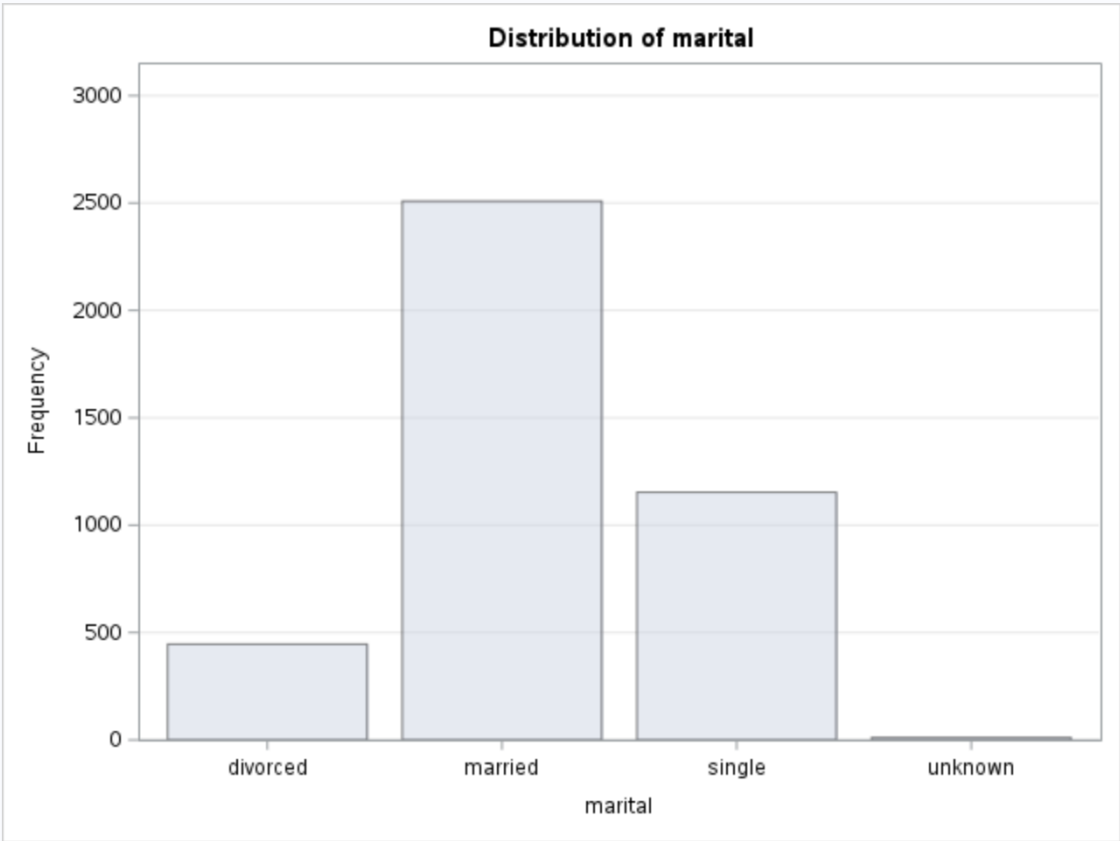
education	Frequency	Percent	Cumulative Frequency	Cumulative Percent
basic.4y	429	10.42	429	10.42
basic.6y	228	5.54	657	15.95
basic.9y	574	13.94	1231	29.89
high.school	921	22.36	2152	52.25
illiterate	1	0.02	2153	52.27
professional.course	535	12.99	2688	65.26
university.degree	1264	30.69	3952	95.95
unknown	167	4.05	4119	100.00

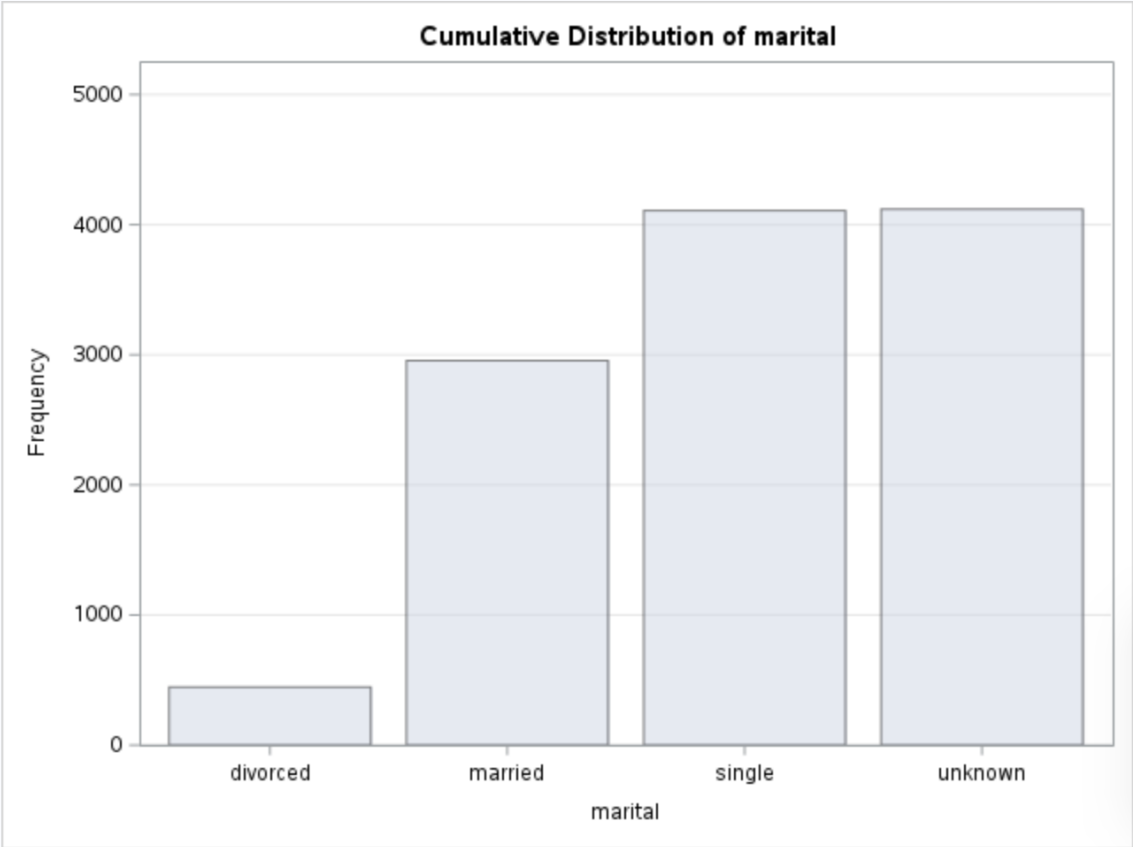




- Bar Chart of Marital Status

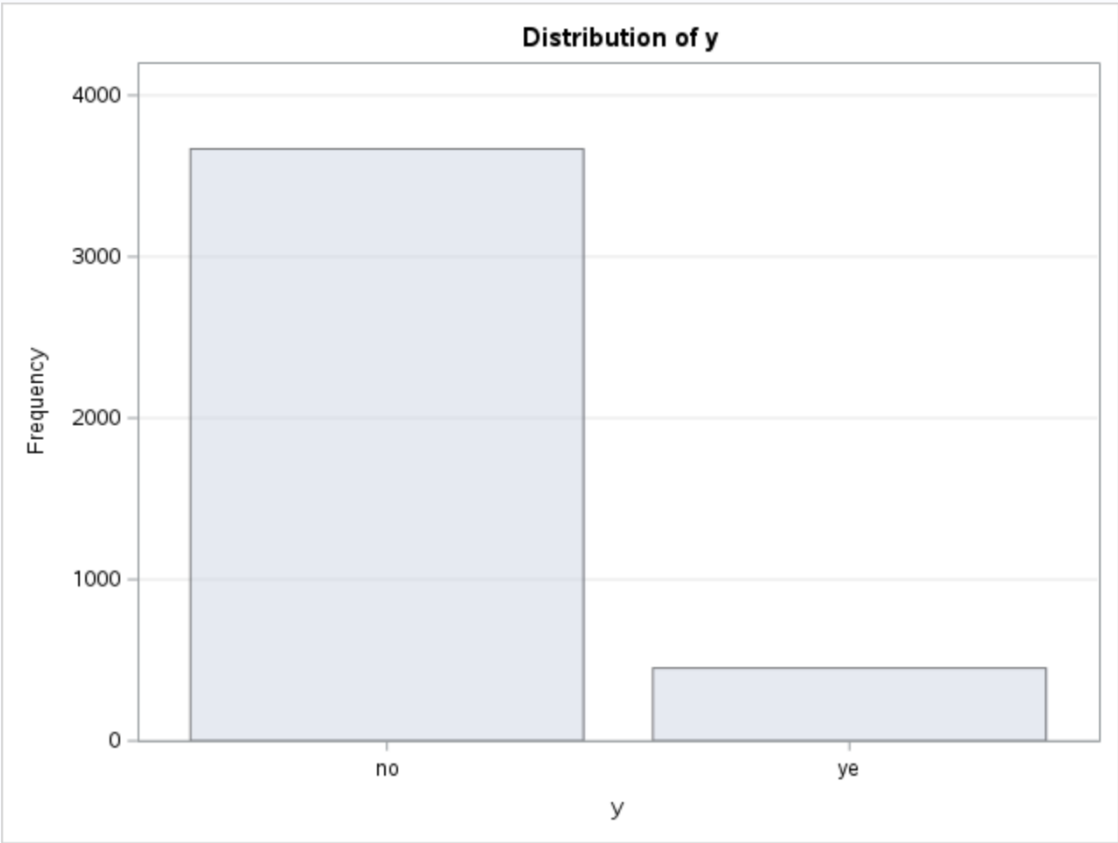
marital	Frequency	Percent	Cumulative Frequency	Cumulative Percent
divorced	446	10.83	446	10.83
married	2509	60.91	2955	71.74
single	1153	27.99	4108	99.73
unknown	11	0.27	4119	100.00

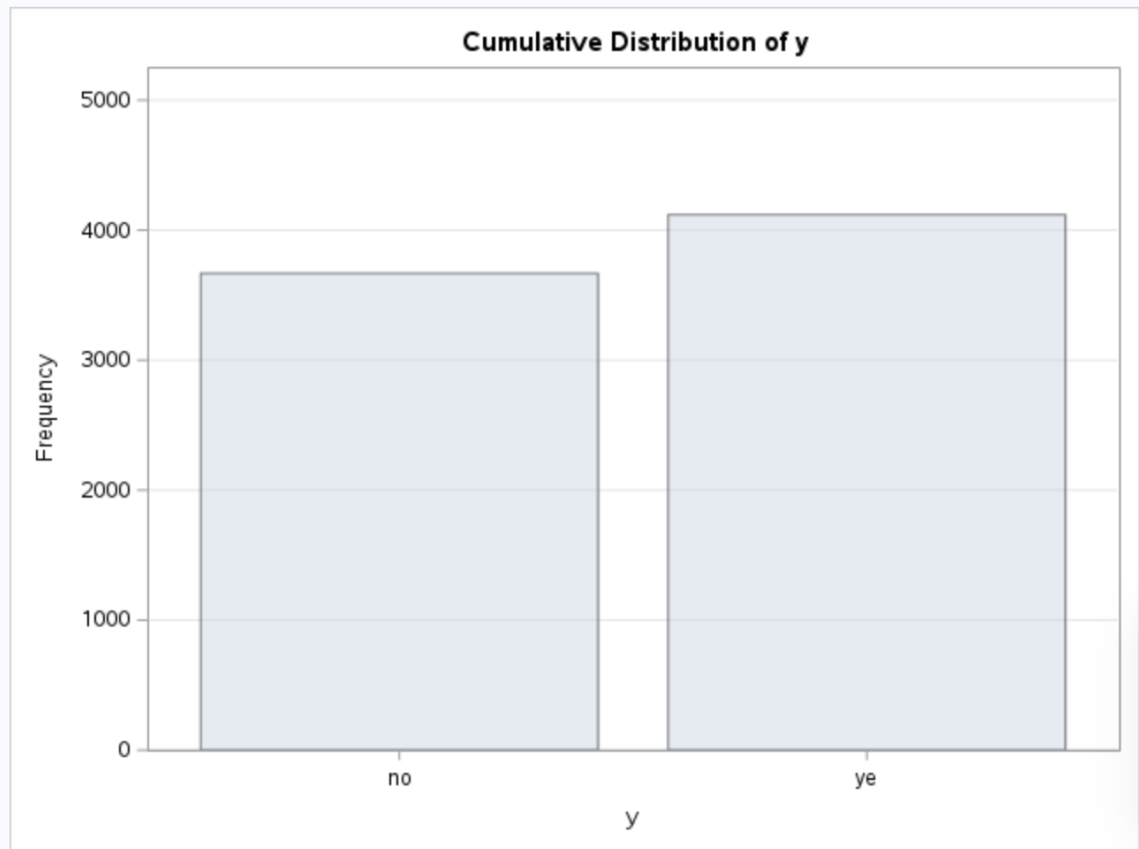




- Bar Chart of Variable y (Subscription)

y	Frequency	Percent	Cumulative Frequency	Cumulative Percent
no	3668	89.05	3668	89.05
ye	451	10.95	4119	100.00





Key Findings

The graphical summaries show that most customers are married, many have a university degree, and administrative and blue-collar occupations are the largest groups. The target variable is highly imbalanced because nearly 89% of customers did not subscribe to a term deposit, while only about 11% accepted the offer.

Appropriate Classification Technique

Because the target variable (y) has only two possible outcomes (Yes or No), this is a supervised classification problem. A Decision Tree is an appropriate technique because it handles both numerical and categorical variables, is easy to interpret, and produces decision rules that can help marketing managers identify customers who are more likely to subscribe to a term deposit.

Conclusion

The exploratory analysis demonstrates that the dataset is suitable for predictive modeling. The descriptive statistics and graphs provide insight into customer characteristics and reveal an imbalanced target variable. A Decision Tree is recommended for the next stage of the project.

References

Moro, S., Cortez, P., & Rita, P. (2014). *A data-driven approach to predict the success of bank telemarketing*. *Decision Support Systems*, 62, 22–31.